
1. Find the asymptotes of $y = \frac{4x^2 + 8}{2x + 1}$.

2. Sketch a graph of the following. Label all axes intercepts with their coordinates and all asymptotes with their equation.

$$f(x) = \frac{x^3 - 3x^2 + 2}{x^2 - 4}$$

3. Consider the function $h : (-\infty, 0) \rightarrow \mathbb{R}$, $h(x) = \frac{x^3 + 1}{2x^2}$ and let h^{-1} be the inverse function of h .

(a) Sketch a graph of $y = h^{-1}(x)$ over its maximal domain. Label all axes intercepts with their coordinates and all asymptotes with their equation.

(b) Find the value of $h^{-1}(1)$.

4. Find the equation of any asymptotes of the following function:

$$j(x) = \left| \frac{3 - 2x^2}{2x - 3} \right|$$